

Claudio César Claros Olivares

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Education

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|---|--|
| M.Sc. Universidad Mayor de San Andrés (UMSA), Electronics Engineering | La Paz, Bolivia
Oct 2020 – present |
| M.Sc. University of Delaware, Electrical and Computer Engineering <ul style="list-style-type: none"> • Concentration: Signal Processing, Communications, and Controls • Thesis: <i>Synergistic Human-Machine Prediction: Active Error Analysis and Mitigation with Gaussian Process Regression</i> (advisor: Austin J. Brockmeier) • GPA: 3.93/4.00 | Newark, DE, USA
Aug 2018 – Aug 2020 |
| Certificate Centro Psicopedagógico y de Investigación en Educación Superior (CEPIES), University Teaching | La Paz, Bolivia
Aug 2015 – Nov 2015 |
| B.S. Universidad Mayor de San Andrés (UMSA), Electronics Engineering <ul style="list-style-type: none"> • Major: Control Systems • Graduation project: <i>Prototipo para generación de consignas de control a partir de los ritmos cerebrales alpha y mu. Caso de estudio: Factibilidad de uso de la plataforma Emotiv EPOC</i> (advisor: Javier Sanabria García) • Class valedictorian | La Paz, Bolivia
Feb 2007 – Dec 2014 |

Experience

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| Computational Neural and Information Engineering Lab, University of Delaware, Research Assistant <ul style="list-style-type: none"> • Developed a synergistic system for human-machine prediction through active error analysis of a model's systematic errors using Gaussian process regression. | Newark, DE, USA
July 2019 – Aug 2020
1 year 2 months |
| Institute of Applied Electronics, UMSA, Researcher <ul style="list-style-type: none"> • Designed and implemented embedded software for signal acquisition and control of a mobile robot prototype for multi-agent research. | La Paz, Bolivia
June 2017 – Dec 2017
7 months |
| Institute of Applied Electronics, UMSA, Researcher <ul style="list-style-type: none"> • Developed a platform of mobile sensor nodes for controlled environments, covering the design of the robotic units and the data acquisition and signal processing tools. | La Paz, Bolivia
Apr 2016 – Dec 2016
9 months |
| Agencia para el Desarrollo de la Sociedad de la Información en Bolivia (ADSIB), Research Intern <ul style="list-style-type: none"> • Designed and implemented an FPGA interface for Ethernet communication dedicated to packet acquisition and routing for the FPGA-based router project. | La Paz, Bolivia
Sept 2015 – Dec 2015
4 months |
| Digital Signal Processing Lab, Electronics Engineering Department, UMSA, Laboratory Assistant <ul style="list-style-type: none"> • Responsible for the maintenance and operation of the digital signal processing laboratory. | La Paz, Bolivia
July 2013 – Dec 2013
6 months |
| Institute of Applied Electronics, UMSA, Research Assistant <ul style="list-style-type: none"> • Developed a software application for EEG acquisition and spectral analysis via the fast Fourier transform. | La Paz, Bolivia
Mar 2012 – Aug 2012
6 months |

Publications

Max-sliced Bures distance for interpreting discrepancies

Jan 2021

Austin J. Brockmeier, Claudio César Claros Olivares, Matthew S. Emigh, Luis G. Sanchez Giraldo
(Preprint)

Synergistic human-machine prediction: Active error analysis and mitigation with Gaussian process regression

Jan 2020

Claudio César Claros Olivares
(M.Sc. thesis, University of Delaware)

Prototipo para generación de consignas de control a partir de los ritmos sensomotores. Caso de estudio: Factibilidad de empleo de la plataforma Emotiv Epoc

Jan 2015

Claudio César Claros Olivares, Javier Sanabria García
(IEEE Bolivian Engineering and Technology Congress (BETCON), La Paz, Bolivia)

Presentations

- **Feb. 2020:** Poster, *Active error analysis and mitigation for synergistic machine learning infrastructure*, DARWIN Computing Symposium, Data Science Institute, University of Delaware.
- **Nov. 2019:** Invited talk, *Learning an error-aware model for synergistic human-machine prediction*, SIG-AI Seminar, Department of Computer and Information Sciences, University of Delaware.
- **Apr. 2016:** Invited talk, *Digital signal processing algorithms and machine learning*, IEEE Seminar on Artificial Intelligence, IEEE Student Branch, Universidad Católica Boliviana, La Paz.
- **May 2015:** Invited talk on the *Sensor your Swarm* project, First International Scientific Congress on Engineering, La Paz.
- **Aug. 2014:** Invited talk, *Research on brain signal processing*, La Paz Ciudad Digital 2.0, representing the Institute of Applied Electronics, UMSA.

Honors & Awards

- **2017:** Bolivian Government Scholarship for graduate studies abroad (30 of 100 scholarships awarded nationwide that year).
- **2015:** Honorable Mention, *Most Challenging Team*, Minesweepers: Towards a Landmine-Free World 2015, Antofagasta, Chile (Bolivian Team).
- **2015:** Senatorial Declaration, Chamber of Senators of the Plurinational Legislative Assembly of Bolivia, for outstanding participation in the NASA Space Apps Challenge 2015.
- **2015:** Resolution of the Honorable University Council, UMSA, for outstanding participation in the NASA Space Apps Challenge 2015.
- **2015:** Worldwide finalist, NASA Space Apps Challenge 2015: nationwide winner for *Best Robotics Theme Project* and worldwide finalist for *Best Mission Concept Project* with the project *Sensor your Swarm*.
- **2015:** Maximum score on the undergraduate graduation project, Electronics Engineering Department, UMSA.
- **2010:** Remarkable academic performance, UMSA. Congratulatory letter awarded to the best students of the year.

Technical Skills

Programming languages: Python, C/C++, MATLAB, Octave, JavaScript, Julia

Libraries: OpenCV, CUDA, SciPy, NumPy, pandas

Machine learning frameworks: TensorFlow, PyTorch, Keras, scikit-learn

Languages

Spanish

Native

English

Fluent (TOEFL 104/120, ECCE)

French

Intermediate (DEL F B2)